Linear Regression 秋 2026
Homework 1

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A is singular $\Leftrightarrow |\operatorname{Var} X| = 0 \Leftrightarrow |a \operatorname{Var} X a^\top| = 0 \Leftrightarrow |\operatorname{Var} a^\top X| = 0 \Leftrightarrow a^\top X = b$

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$$\mathbb{E}X_1^2 = 1 + \mu^2 \quad \mathbb{E}X_1X_2 = 0 + \mu^2$$
$$\mathbb{E}X_2X_3 = \frac{1}{4} + \mu^2 \quad \mathbb{E}X_3^2 = 1 + \mu^2$$

$\mathbb{E}[X_1^2 + 2X_1X_2 - 4X_2X_3 + X_3^2] = 1$

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i

$$\begin{aligned}\operatorname{Var}((1\ \ -2\ \ 3)X) &= (1\ \ -2\ \ 3)\operatorname{Var} X\begin{pmatrix}1\\-2\\3\end{pmatrix}\\ &= (1\ \ -2\ \ 3)\begin{pmatrix}5&2&3\\2&3&0\\3&0&3\end{pmatrix}\begin{pmatrix}1\\-2\\3\end{pmatrix}\\ &= 54\end{aligned}$$

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$$\begin{aligned}\operatorname{Var} Y &= \begin{pmatrix}1&1&0\\1&1&1\end{pmatrix}\begin{pmatrix}5&2&3\\2&3&0\\3&0&3\end{pmatrix}\begin{pmatrix}1&1\\1&1\\0&1\end{pmatrix}\\ &= \begin{pmatrix}12&15\\15&21\end{pmatrix}\end{aligned}$$

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$$\begin{aligned}\operatorname{Var} X &= \mathbb{E}XX^\top - \mathbb{E}X\mathbb{E}X^\top\\ &= \mathbb{E}\big[\mathbb{E}XX^\top|Y\big] - \mathbb{E}[\mathbb{E}X|Y]\mathbb{E}\big[\mathbb{E}X^\top|Y\big]\\ &= \mathbb{E}\big[\mathbb{E}XX^\top|Y\big] - \mathbb{E}\big[\mathbb{E}X|Y\mathbb{E}X^\top|Y\big] + \mathbb{E}\big[\mathbb{E}X|Y\mathbb{E}X^\top|Y\big] - \mathbb{E}[\mathbb{E}X|Y]\mathbb{E}\big[\mathbb{E}X^\top|Y\big]\\ &= \operatorname{Var}(\mathbb{E}X|Y) + \mathbb{E}\operatorname{Var}(X|Y)\end{aligned}$$

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let $Z = AY$

$$\begin{aligned}\mathbb{E}X^\top AY = \mathbb{E}X^\top Z &= \operatorname{Cov}(X^\top, Z^\top) + \mathbb{E}X^\top \mathbb{E}Z \\ &= A\operatorname{Cov}(Y, X) + \mathbb{E}X^\top A\mathbb{E}Y\end{aligned}$$

get trace, then $\mathbb{E}X^\top AY = \operatorname{tr}(A\operatorname{Cov}(Y,X)) + \mathbb{E}X^\top A\mathbb{E}Y$

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$$X = \begin{pmatrix} 1 & 0 & 0 & \dots & 0 \\ 1 & 1 & 0 & \dots & 0 \\ \vdots & & & & \\ 1 & 1 & 1 & \dots & 1 \end{pmatrix} Y$$

thus,

$$\begin{aligned}\operatorname{Var} X &= \begin{pmatrix} 1 & 0 & 0 & \dots & 0 \\ 1 & 1 & 0 & \dots & 0 \\ \vdots & & & & \\ 1 & 1 & 1 & \dots & 1 \end{pmatrix} \operatorname{id}_{n\times n} \begin{pmatrix} 1 & 0 & 0 & \dots & 0 \\ 1 & 1 & 0 & \dots & 0 \\ \vdots & & & & \\ 1 & 1 & 1 & \dots & 1 \end{pmatrix}^\top \\ &= \begin{pmatrix} 1 & 1 & 1 & \dots & 1 \\ 1 & 2 & 2 & \dots & 2 \\ \vdots & & & & \\ 1 & 2 & 3 & \dots & n \end{pmatrix}\end{aligned}$$

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$$\begin{aligned}\operatorname{Cov}(X^\top AX, X^\top BX) &= \mathbb{E}[X^\top AXX^\top BX] - \mathbb{E}X^\top AX\mathbb{E}X^\top BX \\ &= \mathbb{E}[\operatorname{tr}(X^\top AXX^\top BX)] - \sigma^2 \operatorname{tr} A\sigma^2 \operatorname{tr} B \\ &= \operatorname{tr}(A\mathbb{E}[X^\top XBXX^\top]) - \sigma^4 \operatorname{tr} A \operatorname{tr} B \\ &= \operatorname{tr}(A\sigma^4(\operatorname{tr} B + 2B)) - \sigma^4 \operatorname{tr} A \operatorname{tr} B \\ &= 2\sigma^4 \operatorname{tr}(AB)\end{aligned}$$